A6B2_EXPT-8: BLACK BOX TESTING

Test Case Document - Unified Student Placement System (Website)

Particular	Details
Course	Software Engineering Lab (24CS01PR0502)
Experiment	8 - Black Box Testing
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Class / Batch	5S CSE / A6 - B2
Roll No.	28
Date of test execution	26-10-2026
Application under test	Unified Student Placement System (USPS) - placement portal (4 form interfaces)
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Aim

To compile a test case document for a website having at least three form interfaces (excluding Login/Register forms) in a prescribed template, indicating the inputs and the outputs of every test case.

Problem Statement / Application Under Test

The Unified Student Placement System (USPS) is the case study of my SE lab work (SRS, use cases, class, sequence, statechart and activity diagrams). It brings the campus placement process onto one website: recruiters post jobs with eligibility criteria, the placement officer approves them, the system checks each student against the criteria automatically, and students apply and track their applications. The website under test is the USPS prototype built for my SE PBL; its page has a Student, a Recruiter and a Placement Officer view and four form interfaces, none of which is a Login or Register form:

- Form 1 - Post a Job (Recruiter view): A recruiter of an approved company posts a job with its package, deadline and eligibility criteria; it is stored as Pending for the placement cell (SRS FR-09, FR-10, FR-13).
- Form 2 - Record Job Decision (Placement Officer view): The placement officer approves, rejects or returns a pending job posting; on approval the eligible students are notified (SRS FR-11, BR-02).
- Form 3 - Apply for a Job (Student view): A student applies to a published job with one of their resumes and an optional cover note and receives a tracking id (SRS FR-15, UC-1, BR-03, BR-04, BR-05).
- Form 4 - Job Search (Student view (Job board)): A student searches the published jobs by keyword, location and minimum package and can show only the jobs they are eligible for (SRS FR-12, FR-14).

Black Box Testing - Techniques Used

Black box testing checks the software against its specification without looking at the internal code. The tester supplies inputs through the user interface and compares the output shown with the expected output.

- Equivalence Partitioning (EP): each input is divided into valid and invalid classes, and one representative value from each class is tested.
- Boundary Value Analysis (BVA): values at and just outside the limits (min-1, min, max, max+1) are tested, because errors are most likely at the edges of a range.

Test Environment

Item	Details
Website	Unified Student Placement System (HTML, CSS, JavaScript page; Python 3.11 REST server with SQLite), served locally from ../PBL_Student_Placement
Browser	Microsoft Edge 156.0.0.0 (headless) on Windows 11
Execution	Each test case was executed through the user interface: the page is opened as the user of the test case, the input data is entered in the fields, the button is clicked and the message displayed on the page is recorded as the actual output (harness/case.html, run_tests.py).
Test data	The server is started afresh with the pre-loaded data of Tables 1-3 for every test case, with its date and time fixed at 26-10-2026 10:00 so the job deadlines are fixed.
Executed on	26-10-2026 12:54:48

Test Data (Pre-loaded Records)

Table 1 - Students and their resumes

ID	Name	Branch	Batch	CGPA	Backlogs	Placed	Resumes
101	Aarav Kulkarni	CSE	2023-2027	8.60	0	No	Aarav_Resume_v2.pdf (active), Aarav_Resume_v1.pdf (inactive)
102	Diya Joshi	IT	2023-2027	7.40	0	No	Diya_Resume.pdf
103	Kabir Deshmukh	ECE	2023-2027	6.80	1	No	Kabir_Resume.pdf
104	Meera Patil	CSE	2023-2027	9.10	0	Yes	Meera_Resume.pdf
105	Rohan Wankhede	ME	2022-2026	7.90	2	No	Rohan_Resume.pdf

Table 2 - Recruiters and companies; placement officer

Recruiter	Company	Company approved
Ritika Menon	Nimbus Analytics Pvt Ltd	Yes
Sameer Khan	Orbit Systems	Yes
Anil Rao	Vertex Labs	No
Placement officer: Dr. N. Bhagat	-	-

Table 3 - Jobs

Job	Title	Company	Package	Location	Deadline	Status	Eligibility criteria
J-1	Software Engineer	Nimbus	8.5 LPA	Pune	25-11-2026 10:00	Published	CGPA >= 7.0, CSE/IT, 0 backlogs, batch 2023-2027
J-2	Embedded Systems Engineer	Orbit	6.0 LPA	Bengaluru	15-11-2026 10:00	Published	CGPA >= 6.5, CSE/ECE/EE, <= 1 backlog
J-3	Data Analyst	Nimbus	7.0 LPA	Hyderabad	20-11-2026 10:00	Pending	CGPA >= 7.5, 0 backlogs
J-4	Firmware Intern	Orbit	3.6 LPA	Nagpur	25-10-2026 10:00 (passed)	Published	CGPA >= 6.0, ECE/EE

Form Interfaces - Inputs and Outputs

Form 1: Post a Job (Recruiter view)

Input field	Control	Valid input	Output (message) for invalid input
Job title	Text box	3 to 120 characters	Title must be 3 to 120 characters.
Description	Text area	20 to 2000 characters	Description must be 20 to 2000 characters.
Package (LPA)	Text box	A number, more than 0 and at most 100	Package (CTC) must be a positive number up to 100 LPA. / Package (CTC) must be a number.
Location	Text box	2 to 80 characters	Location is required (2 to 80 characters).
Application deadline	Date and time	Later than the current date and time	Deadline must be a future date and time.
Minimum CGPA	Text box	Optional; 0 to 10	Minimum CGPA must be between 0 and 10.
Maximum active backlogs	Text box	Optional; whole number 0 to 10	Maximum backlogs must be a whole number from 0 to 10.
Allowed branches	Check boxes	Optional; CSE, IT, ECE, EE, ME, CE	-
Batch	Text box	Optional; YYYY-YYYY spanning 4 years (e.g. 2023-2027)	Batch must look like 2023-2027.
Eligibility criteria (as a group)	Rule	At least one of the four criteria	At least one eligibility criterion is required.
Company of the recruiter	Rule	Approved by the placement cell	Your company is not yet approved by the placement cell.

Output for valid input: Job J-<id> submitted for approval (status: Pending). The job appears in My job postings as Pending.

Form 2: Record Job Decision (Placement Officer view)

Input field	Control	Valid input	Output (message) for invalid input
Job	Drop-down	A pending job posting	Select a pending job posting.
Decision	Radio buttons	Approve, Reject or Return for changes	Decision must be approve, reject or return.
Remarks	Text box	Reject / Return: 5 to 200 characters; Approve: optional, at most 200	Remarks (5 to 200 characters) are required to reject or return a posting. / Remarks cannot exceed 200 characters.

Output for valid input: Job J-<id> published. <n> eligible student(s) notified. / Job J-<id> rejected. Recruiter notified. / Job J-<id> returned to the recruiter for changes.

Form 3: Apply for a Job (Student view)

Input field	Control	Valid input	Output (message) for invalid input
Job	Drop-down	A published job the student is eligible for, before its deadline	Select a job to apply for. / Not eligible: <unmet criteria> (Submit disabled) / The application deadline for J-<id> has passed. (Submit disabled) / You are already placed; the placement policy blocks further applications. / You have already applied to J-<id> (tracking id <tid>).
Resume	Drop-down	One of the student's active resumes (the active one is pre-selected)	Select one of your active resumes.
Cover note	Text area	Optional; at most 500 characters (counter shown)	Cover note cannot exceed 500 characters.

Output for valid input: Application submitted for J-<id>. Tracking id USPS-26-<nnnn>.

Form 4: Job Search (Student view (Job board))

Input field	Control	Valid input	Output (message) for invalid input
Keyword	Text box	Any text; matches job title, description or company, ignoring case	-
Location	Text box	Any text; part of the job location, ignoring case	-
Min package (LPA)	Text box	Optional; a number	Minimum package must be a number.
Eligible only	Check box	Ticked or not	-

Output for valid input: Job count and the matching jobs, open jobs first: '<n> jobs' followed by the job titles; when nothing matches: 'No jobs match the filters.'

Test Case Document

Template: TC ID | Test scenario (technique) | Pre-condition | Test steps | Input (test data) | Expected output | Actual output | Status. Dates are shown as dd-mm-yyyy; the test cases were executed on 26-10-2026.

Form 1: Post a Job

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_PJ_01	All fields valid [EP (valid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Job title: Backend Developer Description: Design REST APIs and database schemas for the analytics platform. (65 chars) Package (LPA): 9 Location: Pune Deadline: 05-11-2026 17:00 Minimum CGPA: 7 Maximum backlogs: 0 Branches: CSE, IT Batch: (blank)	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**
TC_PJ_02	Job title of 2 characters [BVA (min - 1)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Job title: QA (2 chars) Other fields: as in TC_PJ_01	Title must be 3 to 120 characters.	Title must be 3 to 120 characters.	**PASS**
TC_PJ_03	Job title of 3 characters [BVA (min)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Job title: SDE (3 chars) Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**
TC_PJ_04	Job title of 120 characters [BVA (max)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Job title: 'A' x 120 Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_PJ_05	Job title of 121 characters [BVA (max + 1)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Job title: 'A' x 121 Other fields: as in TC_PJ_01	Title must be 3 to 120 characters.	Title must be 3 to 120 characters.	**PASS**
TC_PJ_06	Description of 19 characters [BVA (min - 1)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Description: Write code for APIs (19 chars) Other fields: as in TC_PJ_01	Description must be 20 to 2000 characters.	Description must be 20 to 2000 characters.	**PASS**
TC_PJ_07	Description of 20 characters [BVA (min)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Description: Write code for APIs. (20 chars) Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**
TC_PJ_08	Package of 0 LPA [BVA (min - 1)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Package (LPA): 0 Other fields: as in TC_PJ_01	Package (CTC) must be a positive number up to 100 LPA.	Package (CTC) must be a positive number up to 100 LPA.	**PASS**
TC_PJ_09	Package just above 0 [BVA (min)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Package (LPA): 0.1 Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**
TC_PJ_10	Package of 100 LPA [BVA (max)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Package (LPA): 100 Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_PJ_11	Package above 100 LPA [BVA (max + 0.5)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Package (LPA): 100.5 Other fields: as in TC_PJ_01	Package (CTC) must be a positive number up to 100 LPA.	Package (CTC) must be a positive number up to 100 LPA.	**PASS**
TC_PJ_12	Package typed in words [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Package (LPA): nine Other fields: as in TC_PJ_01	Package (CTC) must be a number.	Package (CTC) must be a number.	**PASS**
TC_PJ_13	Location left blank [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Location: (blank) Other fields: as in TC_PJ_01	Location is required (2 to 80 characters).	Location is required (2 to 80 characters).	**PASS**
TC_PJ_14	Deadline equal to the current date and time [BVA (equal to now)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Deadline: 26-10-2026 10:00 (= now) Other fields: as in TC_PJ_01	Deadline must be a future date and time.	Deadline must be a future date and time.	**PASS**
TC_PJ_15	Deadline one minute in the future [BVA (now + 1 min)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Deadline: 26-10-2026 10:01 Other fields: as in TC_PJ_01	Job J-5 submitted for approval (status: Pending).	Job J-5 submitted for approval (status: Pending).	**PASS**
TC_PJ_16	Deadline not entered [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Deadline: (blank) Other fields: as in TC_PJ_01	Deadline must be a future date and time.	Deadline must be a future date and time.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_PJ_17	Minimum CGPA above 10 [BVA (max + 0.01)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Minimum CGPA: 10.01 Other fields: as in TC_PJ_01	Minimum CGPA must be between 0 and 10.	Minimum CGPA must be between 0 and 10.	**PASS**
TC_PJ_18	Maximum backlogs above 10 [BVA (max + 1)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Maximum backlogs: 11 Other fields: as in TC_PJ_01	Maximum backlogs must be a whole number from 0 to 10.	Maximum backlogs must be a whole number from 0 to 10.	**PASS**
TC_PJ_19	Batch that does not span 4 years [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Batch: 2023-2026 Other fields: as in TC_PJ_01	Batch must look like 2023-2027.	Batch must look like 2023-2027.	**PASS**
TC_PJ_20	No eligibility criterion given [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Ritika Menon - Nimbus Analytics 2. Enter the input data 3. Click 'Submit for approval'	Minimum CGPA, Maximum backlogs, Batch: (blank) Branches: none ticked Other fields: as in TC_PJ_01	At least one eligibility criterion is required.	At least one eligibility criterion is required.	**PASS**
TC_PJ_21	Recruiter of a company not approved by the placement cell [EP (invalid class)]	Recruiter view open as Ritika Menon (Nimbus Analytics, approved) unless stated; data of Tables 1-3; server date 26-10-2026 10:00.	1. Open the Recruiter view as Recruiter Anil Rao - Vertex Labs 2. Enter the input data 3. Click 'Submit for approval'	All fields: as in TC_PJ_01	Your company is not yet approved by the placement cell.	Your company is not yet approved by the placement cell.	**PASS**

Form 2: Record Job Decision

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_JD_01	Approve the pending posting without remarks [EP (valid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Data Analyst - Nimbus Analytics Decision: Approve Remarks: (blank)	Job J-3 published. 1 eligible student(s) notified.	Job J-3 published. 1 eligible student(s) notified.	**PASS**
TC_JD_02	Reject with remarks [EP (valid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Reject Remarks: Package below the campus minimum. (33 chars)	Job J-3 rejected. Recruiter notified.	Job J-3 rejected. Recruiter notified.	**PASS**
TC_JD_03	Return for changes with remarks [EP (valid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Return for changes Remarks: Please add the batch criterion. (31 chars)	Job J-3 returned to the recruiter for changes.	Job J-3 returned to the recruiter for changes.	**PASS**
TC_JD_04	No job selected [EP (invalid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: -- Select -- Decision: Approve Remarks: (blank)	Select a pending job posting.	Select a pending job posting.	**PASS**
TC_JD_05	No decision selected [EP (invalid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: (none) Remarks: (blank)	Decision must be approve, reject or return.	Decision must be approve, reject or return.	**PASS**
TC_JD_06	Reject without remarks [EP (invalid class)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Reject Remarks: (blank)	Remarks (5 to 200 characters) are required to reject or return a posting.	Remarks (5 to 200 characters) are required to reject or return a posting.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_JD_07	Reject with remarks of 4 characters [BVA (min - 1)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Reject Remarks: Late (4 chars)	Remarks (5 to 200 characters) are required to reject or return a posting.	Remarks (5 to 200 characters) are required to reject or return a posting.	**PASS**
TC_JD_08	Reject with remarks of 5 characters [BVA (min)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Reject Remarks: Spam. (5 chars)	Job J-3 rejected. Recruiter notified.	Job J-3 rejected. Recruiter notified.	**PASS**
TC_JD_09	Return with remarks of 200 characters [BVA (max)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Return for changes Remarks: 'R' x 200	Job J-3 returned to the recruiter for changes.	Job J-3 returned to the recruiter for changes.	**PASS**
TC_JD_10	Return with remarks of 201 characters [BVA (max + 1)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Return for changes Remarks: 'R' x 201	Remarks (5 to 200 characters) are required to reject or return a posting.	Remarks (5 to 200 characters) are required to reject or return a posting.	**PASS**
TC_JD_11	Approve with remarks of 200 characters [BVA (max)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Approve Remarks: 'R' x 200	Job J-3 published. 1 eligible student(s) notified.	Job J-3 published. 1 eligible student(s) notified.	**PASS**
TC_JD_12	Approve with remarks of 201 characters [BVA (max + 1)]	Placement Officer view open as Dr. N. Bhagat; J-3 Data Analyst is the pending posting (Table 3). A confirmation dialog for Reject is answered OK.	1. Open the Placement Officer view as Placement Officer Dr. N. Bhagat 2. Enter the input data 3. Click 'Record decision'	Job: J-3 Decision: Approve Remarks: 'R' x 201	Remarks cannot exceed 200 characters.	Remarks cannot exceed 200 characters.	**PASS**

Form 3: Apply for a Job

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AJ_01	Eligible student applies with the pre-selected active resume [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Software Engineer - Nimbus Analytics Resume: Aarav_Resume_v2.pdf (pre-selected) Cover note: Keen on backend work. (21 chars)	Application submitted for J-1. Tracking id USPS-26-0001.	Application submitted for J-1. Tracking id USPS-26-0001.	**PASS**
TC_AJ_02	Cover note left blank (optional) [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Diya Joshi 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: Diya_Resume.pdf (pre-selected) Cover note: (blank)	Application submitted for J-1. Tracking id USPS-26-0001.	Application submitted for J-1. Tracking id USPS-26-0001.	**PASS**
TC_AJ_03	No job selected [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: -- Select a job -- Resume: Aarav_Resume_v2.pdf Cover note: (blank)	Select a job to apply for.	Select a job to apply for.	**PASS**
TC_AJ_04	Student not eligible for the job [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Kabir Deshmukh 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Software Engineer (CGPA >= 7.0, CSE/IT, 0 backlogs) Resume: Kabir_Resume.pdf Cover note: (blank)	Not eligible: CGPA 6.80 is below the minimum 7.00. Branch ECE is not in the allowed branches (CSE, IT). Active backlogs 1 exceed the maximum 0. | Submit application button disabled	Not eligible: CGPA 6.80 is below the minimum 7.00. Branch ECE is not in the allowed branches (CSE, IT). Active backlogs 1 exceed the maximum 0. | Submit application button disabled	**PASS**
TC_AJ_05	Job whose deadline has passed [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Kabir Deshmukh 2. Enter the input data 3. Click 'Submit application'	Job: J-4 Firmware Intern (deadline 25-10-2026 10:00) Resume: Kabir_Resume.pdf Cover note: (blank)	The application deadline for J-4 has passed. | Submit application button disabled	The application deadline for J-4 has passed. | Submit application button disabled	**PASS**
TC_AJ_06	Student who is already placed [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Meera Patil 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: Meera_Resume.pdf Cover note: (blank)	You are already placed; the placement policy blocks further applications.	You are already placed; the placement policy blocks further applications.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_AJ_07	Inactive resume selected [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: Aarav_Resume_v1.pdf (inactive) Cover note: (blank)	Select one of your active resumes.	Select one of your active resumes.	**PASS**
TC_AJ_08	No resume selected [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: -- Select a resume -- Cover note: (blank)	Select one of your active resumes.	Select one of your active resumes.	**PASS**
TC_AJ_09	Cover note of 500 characters [BVA (max)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: Aarav_Resume_v2.pdf Cover note: 'a' x 500	Application submitted for J-1. Tracking id USPS-26-0001.	Application submitted for J-1. Tracking id USPS-26-0001.	**PASS**
TC_AJ_10	Cover note of 501 characters [BVA (max + 1)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 Resume: Aarav_Resume_v2.pdf Cover note: 'a' x 501	Cover note cannot exceed 500 characters.	Cover note cannot exceed 500 characters.	**PASS**
TC_AJ_11	Applying to the same job a second time [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Submit application'	Job: J-1 (Submit pressed twice) Resume: Aarav_Resume_v2.pdf Cover note: (blank)	You have already applied to J-1 (tracking id USPS-26-0001).	You have already applied to J-1 (tracking id USPS-26-0001).	**PASS**

Form 4: Job Search

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_JS_01	No filter: all published jobs, open ones first [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Keyword, Location, Min package: (blank) Eligible only: unticked	3 jobs: Embedded Systems Engineer, Software Engineer, Firmware Intern	3 jobs: Embedded Systems Engineer, Software Engineer, Firmware Intern	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_JS_02	Keyword in the job title [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Keyword: software	1 job: Software Engineer	1 job: Software Engineer	**PASS**
TC_JS_03	Keyword in the company name, upper case [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Keyword: ORBIT	2 jobs: Embedded Systems Engineer, Firmware Intern	2 jobs: Embedded Systems Engineer, Firmware Intern	**PASS**
TC_JS_04	Keyword that matches no job [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Keyword: doctor	0 jobs: No jobs match the filters.	0 jobs: No jobs match the filters.	**PASS**
TC_JS_05	Location filter [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Location: pune	1 job: Software Engineer	1 job: Software Engineer	**PASS**
TC_JS_06	Minimum package equal to a job's package (6 LPA) [BVA (equal)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Min package: 6	2 jobs: Embedded Systems Engineer, Software Engineer	2 jobs: Embedded Systems Engineer, Software Engineer	**PASS**
TC_JS_07	Minimum package just above 6 LPA [BVA (equal + 0.1)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Min package: 6.1	1 job: Software Engineer	1 job: Software Engineer	**PASS**
TC_JS_08	Minimum package above the highest package (8.5 LPA) [BVA (max + 0.1)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Min package: 8.6	0 jobs: No jobs match the filters.	0 jobs: No jobs match the filters.	**PASS**

TC ID	Test scenario (technique)	Pre-condition	Test steps	Input (test data)	Expected output	Actual output	Status
TC_JS_09	Minimum package typed in words [EP (invalid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Min package: abc	Minimum package must be a number.	Minimum package must be a number.	**PASS**
TC_JS_10	Eligible only - ECE student [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Kabir Deshmukh 2. Enter the input data 3. Click 'Search'	Eligible only: ticked	2 jobs: Embedded Systems Engineer, Firmware Intern	2 jobs: Embedded Systems Engineer, Firmware Intern	**PASS**
TC_JS_11	Eligible only - student eligible for no job [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Rohan Wankhede 2. Enter the input data 3. Click 'Search'	Eligible only: ticked	0 jobs: No jobs match the filters.	0 jobs: No jobs match the filters.	**PASS**
TC_JS_12	Keyword and location together [EP (valid class)]	Student view open as the student named in the test case; data of Tables 1-3.	1. Open the Student view (Job board) as Student Aarav Kulkarni 2. Enter the input data 3. Click 'Search'	Keyword: engineer Location: Bengaluru	1 job: Embedded Systems Engineer	1 job: Embedded Systems Engineer	**PASS**

Test Execution Summary

Form	Test cases executed	EP / BVA cases	Passed	Failed
Form 1: Post a Job	21	7 / 14	21	0
Form 2: Record Job Decision	12	6 / 6	12	0
Form 3: Apply for a Job	11	9 / 2	11	0
Form 4: Job Search	12	9 / 3	12	0
Total	**56**	**31 / 25**	**56**	**0**

Execution Screenshots

Screenshots were captured in Microsoft Edge after each test case was executed. The dark bar at the top is added by the test harness and shows the test case ID, the time of execution, the actual output and the result.

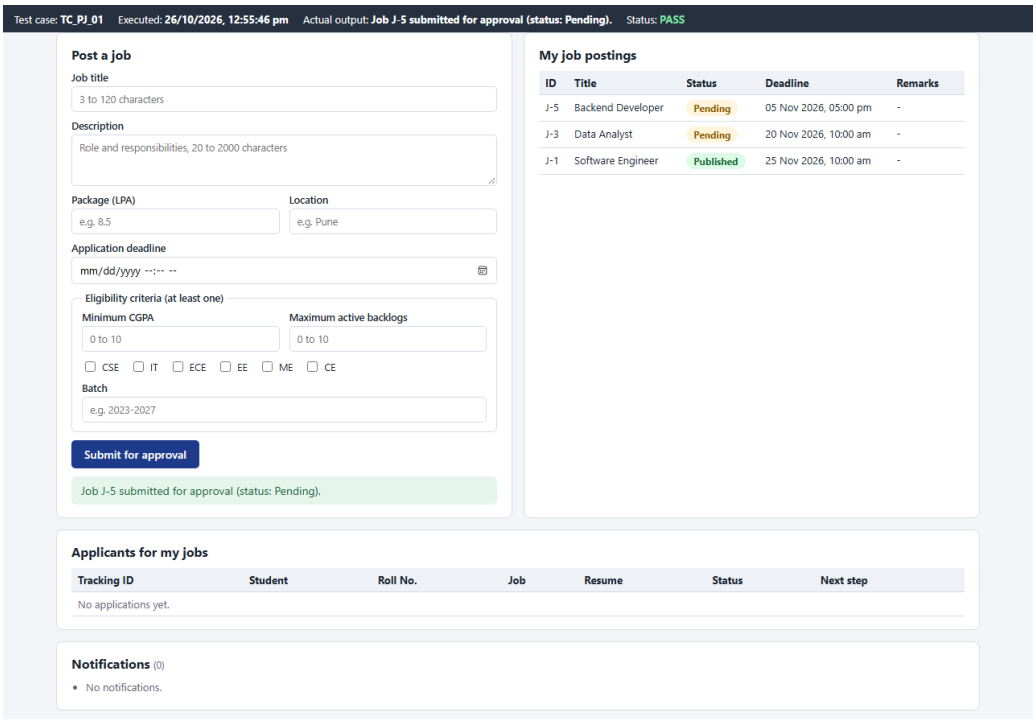
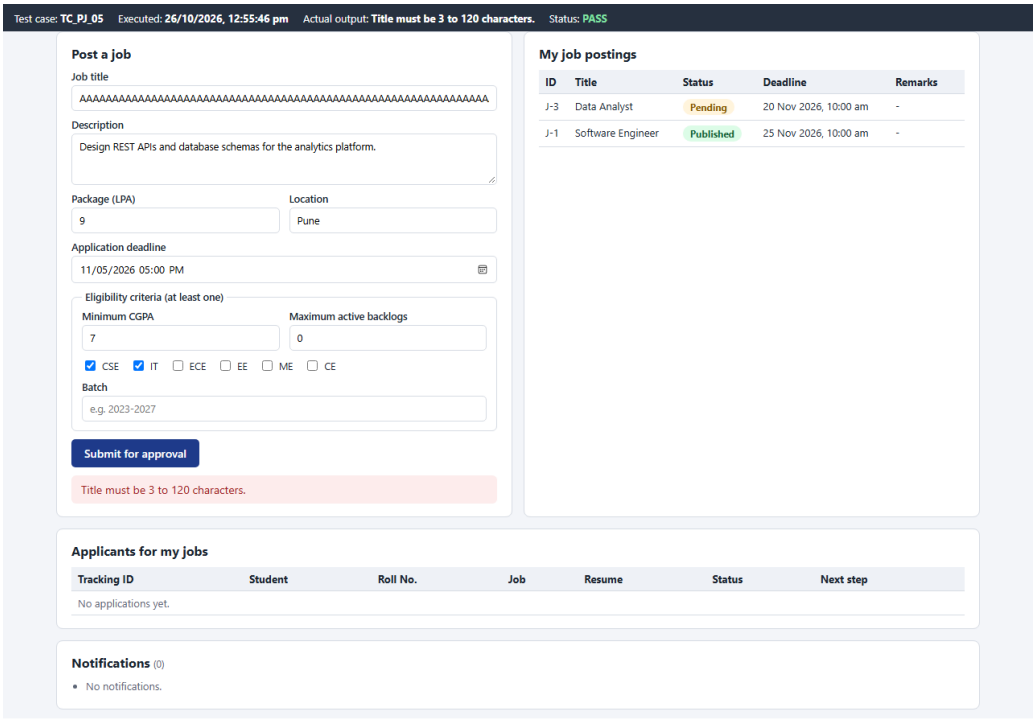


Figure 1: TC_PJ_01 - Form 1 - valid job posting submitted for approval


Test case: TC_PJ_05 Executed: 26/10/2026, 12:55:46 pm Actual output: Title must be 3 to 120 characters. Status: PASS
Post a job
Job title
AAAA

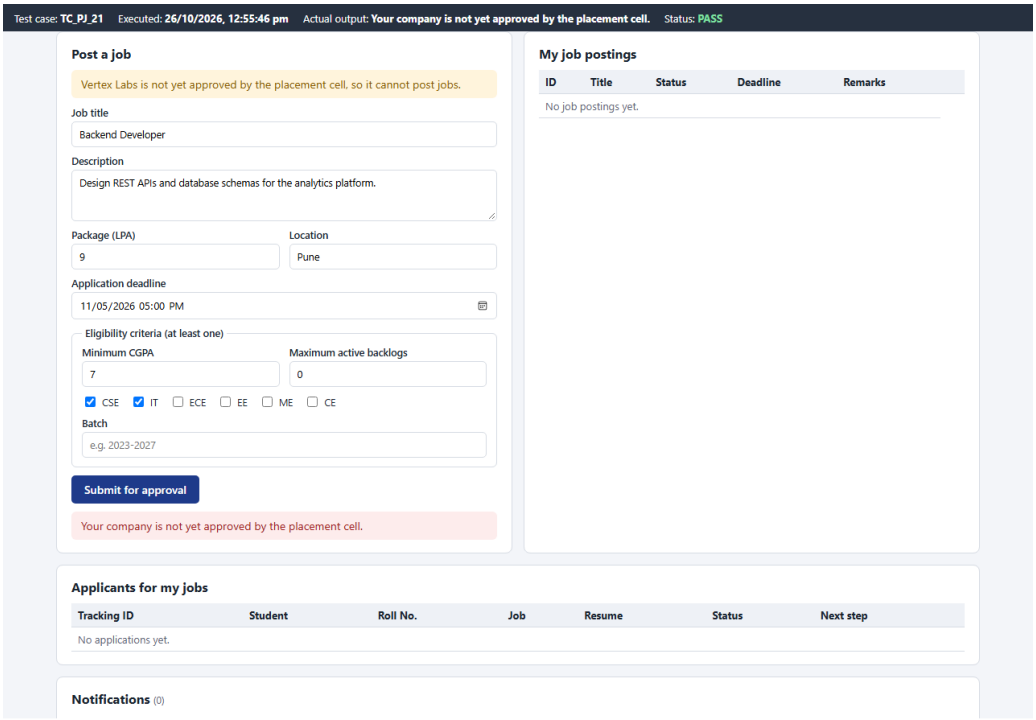


Figure 3: TC_PJ_21 - Form 1 - recruiter of a company that is not approved

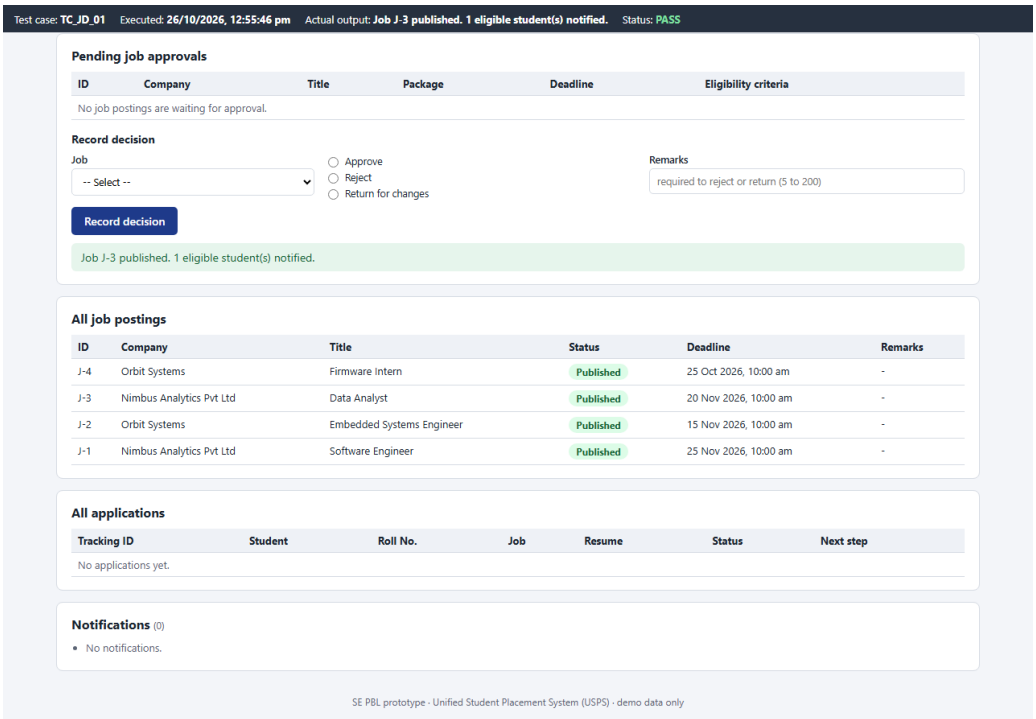


Figure 4: TC_JD_01 - Form 2 - pending job J-3 approved, eligible student notified

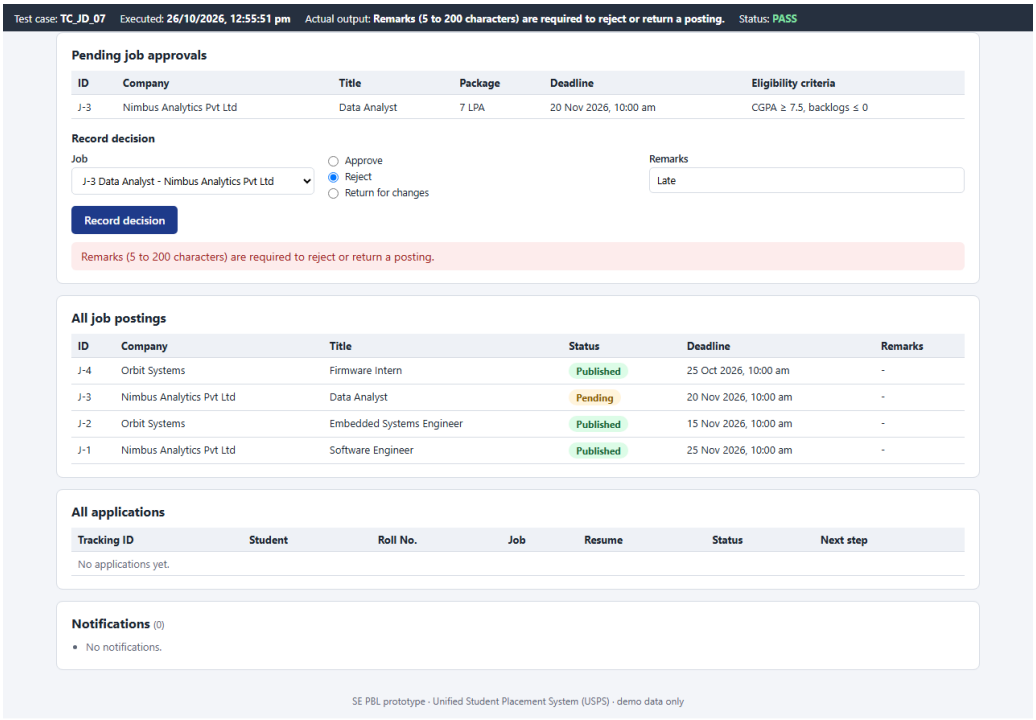


Figure 5: TC_JD_07 - Form 2 - reject with 4-character remarks refused (BVA, min - 1)

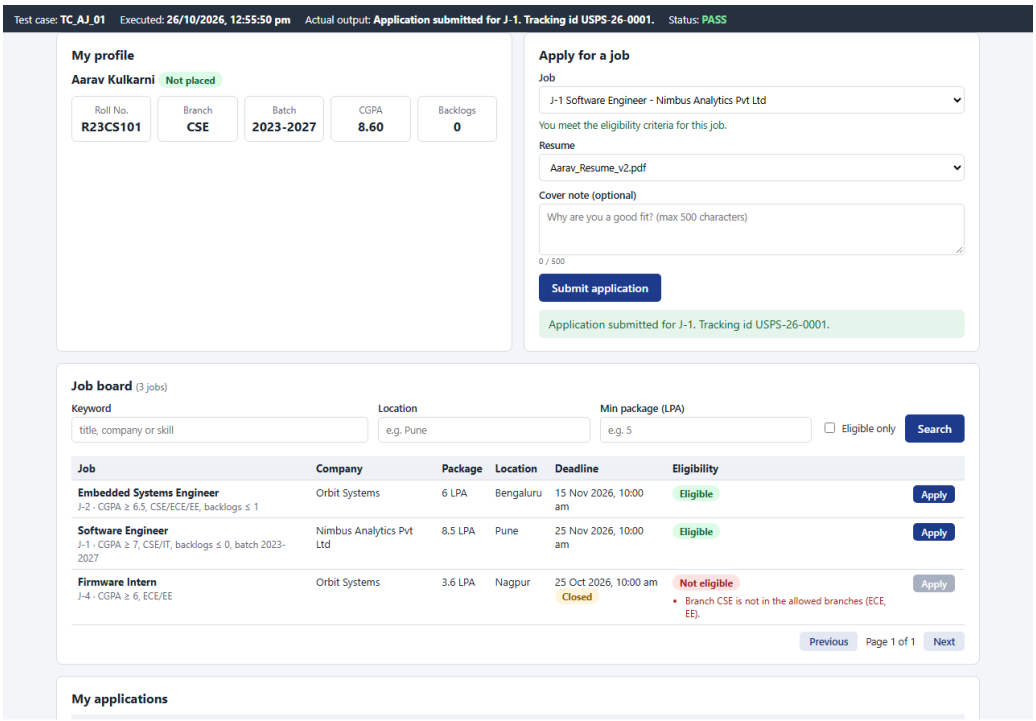


Figure 6: TC_AJ_01 - Form 3 - application submitted with a tracking id

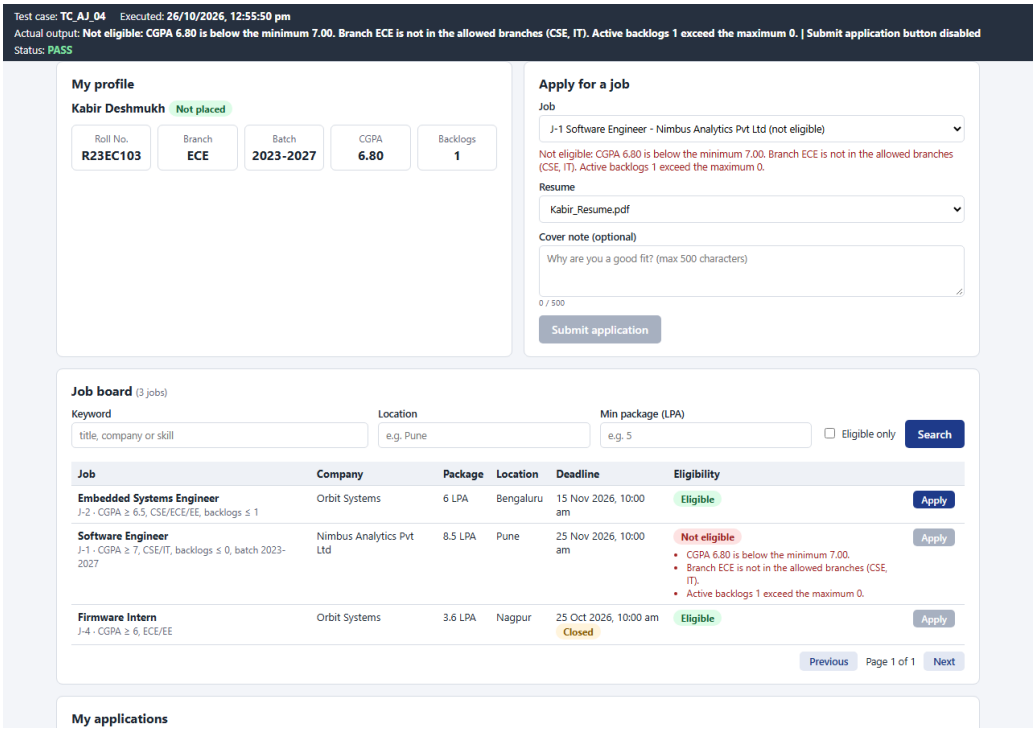


Figure 7: TC_AJ_04 - Form 3 - ineligible student: unmet criteria listed, Submit disabled

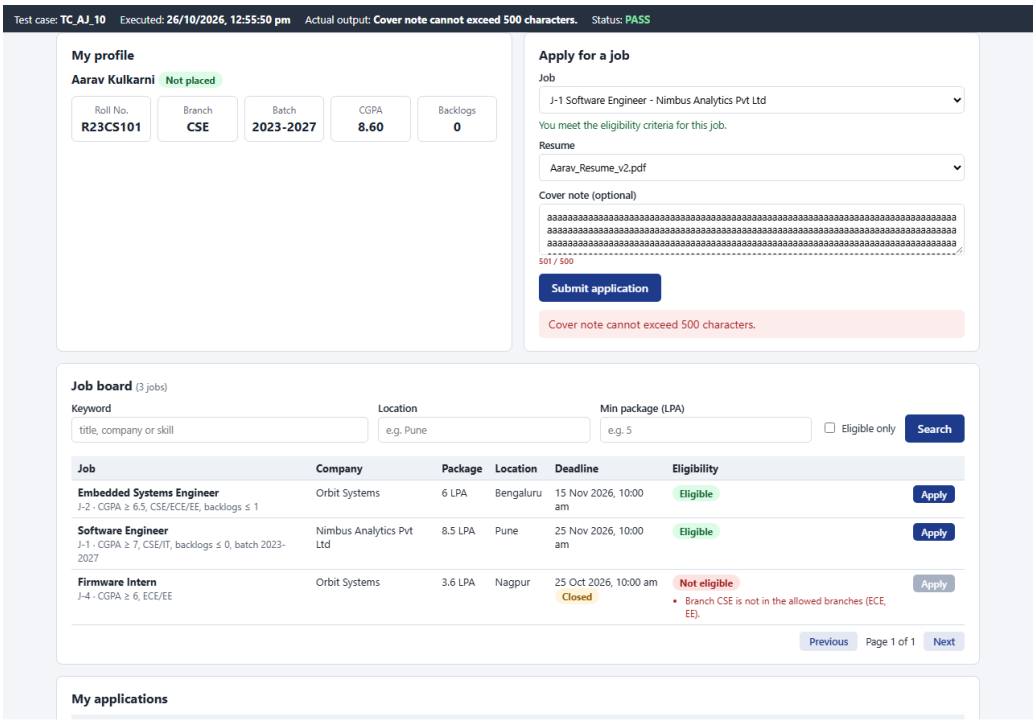


Figure 8: TC_AJ_10 - Form 3 - cover note of 501 characters refused (BVA, max + 1)

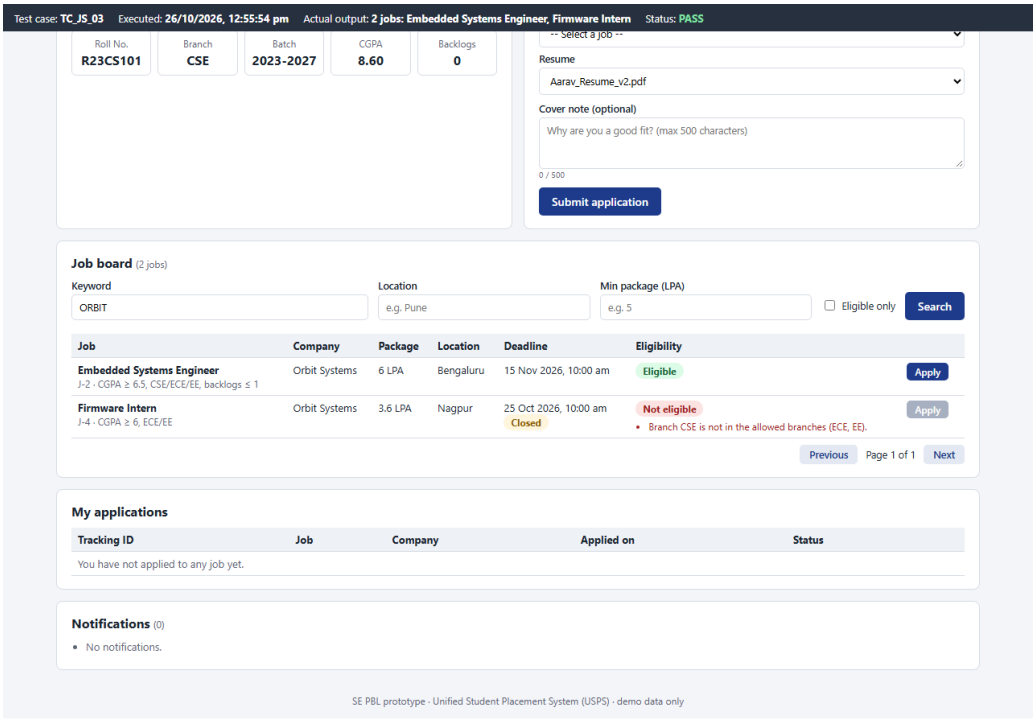


Figure 9: TC_JS_03 - Form 4 - keyword search on the company name

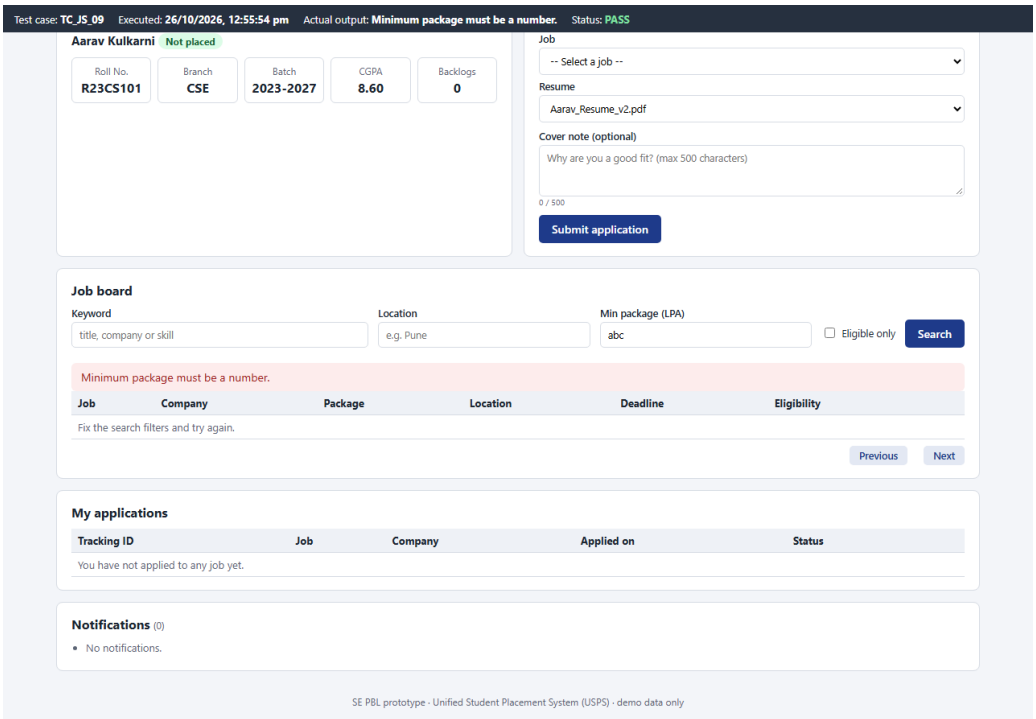


Figure 10: TC_JS_09 - Form 4 - minimum package typed in words

```
Expt-8 black box test run - 26-10-2026 12:54:48

TC_PJ_01  PASS  All fields valid
TC_PJ_02  PASS  Job title of 2 characters
TC_PJ_03  PASS  Job title of 3 characters
TC_PJ_04  PASS  Job title of 120 characters
TC_PJ_05  PASS  Job title of 121 characters
TC_PJ_06  PASS  Description of 19 characters
TC_PJ_07  PASS  Description of 20 characters
TC_PJ_08  PASS  Package of 0 LPA
TC_PJ_09  PASS  Package just above 0
TC_PJ_10  PASS  Package of 100 LPA
TC_PJ_11  PASS  Package above 100 LPA
TC_PJ_12  PASS  Package typed in words
TC_PJ_13  PASS  Location left blank
TC_PJ_14  PASS  Deadline equal to the current date and time
TC_PJ_15  PASS  Deadline one minute in the future
TC_PJ_16  PASS  Deadline not entered
TC_PJ_17  PASS  Minimum CGPA above 10
TC_PJ_18  PASS  Maximum backlogs above 10
TC_PJ_19  PASS  Batch that does not span 4 years
TC_PJ_20  PASS  No eligibility criterion given
TC_PJ_21  PASS  Recruiter of a company not approved by the placement cell
TC_JD_01  PASS  Approve the pending posting without remarks
TC_JD_02  PASS  Reject with remarks
TC_JD_03  PASS  Return for changes with remarks
TC_JD_04  PASS  No job selected
TC_JD_05  PASS  No decision selected
TC_JD_06  PASS  Reject without remarks
TC_JD_07  PASS  Reject with remarks of 4 characters
TC_JD_08  PASS  Reject with remarks of 5 characters
TC_JD_09  PASS  Return with remarks of 200 characters
TC_JD_10  PASS  Return with remarks of 201 characters
TC_JD_11  PASS  Approve with remarks of 200 characters
TC_JD_12  PASS  Approve with remarks of 201 characters
TC_AJ_01  PASS  Eligible student applies with the pre-selected active resume
TC_AJ_02  PASS  Cover note left blank (optional)
TC_AJ_03  PASS  No job selected
TC_AJ_04  PASS  Student not eligible for the job
TC_AJ_05  PASS  Job whose deadline has passed
TC_AJ_06  PASS  Student who is already placed
TC_AJ_07  PASS  Inactive resume selected
TC_AJ_08  PASS  No resume selected
TC_AJ_09  PASS  Cover note of 500 characters
TC_AJ_10  PASS  Cover note of 501 characters
TC_AJ_11  PASS  Applying to the same job a second time
TC_JS_01  PASS  No filter: all published jobs, open ones first
TC_JS_02  PASS  Keyword in the job title
TC_JS_03  PASS  Keyword in the company name, upper case
TC_JS_04  PASS  Keyword that matches no job
TC_JS_05  PASS  Location filter
TC_JS_06  PASS  Minimum package equal to a job's package (6 LPA)
TC_JS_07  PASS  Minimum package just above 6 LPA
TC_JS_08  PASS  Minimum package above the highest package (8.5 LPA)
TC_JS_09  PASS  Minimum package typed in words
TC_JS_10  PASS  Eligible only - ECE student
TC_JS_11  PASS  Eligible only - student eligible for no job
TC_JS_12  PASS  Keyword and location together

56 test cases executed, 56 passed, 0 failed
```

Figure 11: Complete test run (run_tests.py) - 56 test cases executed, 56 passed, 0 failed

Conclusion

A test case document was prepared for four form interfaces of the Unified Student Placement System website: Post a Job, Record Job Decision, Apply for a Job and Job Search. Using Equivalence Partitioning and Boundary Value Analysis, 56 test cases were designed with their inputs and expected outputs. All test cases were executed through the user interface and the actual outputs were recorded: 56 passed and 0 failed.